

# **AI-compute supply chain: chart pack**

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, audited vs Atlas) and Atlas HS2012 annual. Sections: 1 the chain as flows (2024) -- 2 the time-varying factor model -- 3 concentration & fragmentation. Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-07-27

# 1. The chain as flows

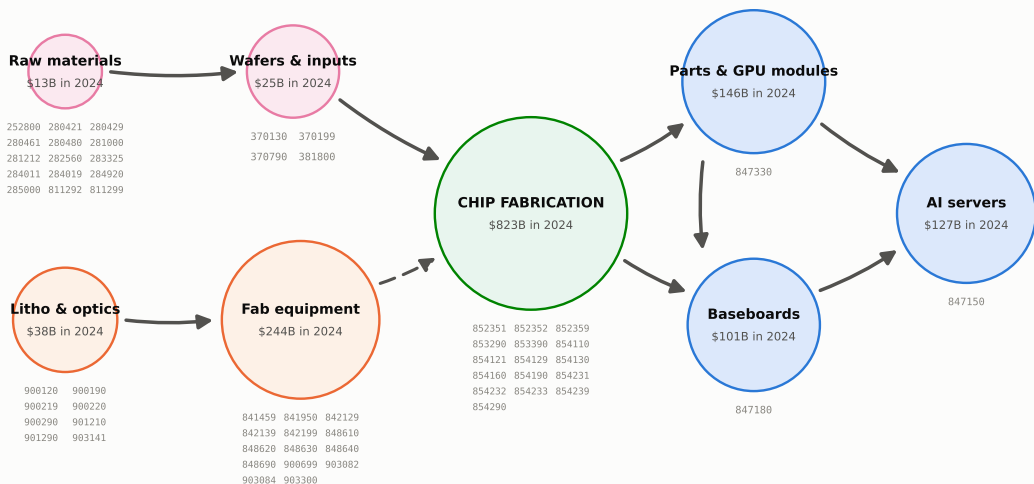
2024 bilateral flows, China+Hong Kong merged (CHK), intra-bloc excluded. Stages 1-5 from Atlas HS2012 annual data; stages 6-8 from the monthly panel (docs/data.md §3).  
Generated by scripts/export\_supply\_chain\_sankey.py.

1. The chain as flows

# The chain's stylized shape, with HS6 codes per node

Two parallel input branches (materials; tools) converge on chip fabrication; the output side is sequential. Node heights scale with the log of 2024 world trade. Solid arrows: consumed per unit of output; dashed: capacity investment (leads output 6-12 months in fab countries).

## The AI-compute supply chain — stylized shape, with the HS6 codes in each node



Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity; in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory is shipped to Taiwan (a flow the data shows) and joins the processors at packaging there; the joining step itself happens inside Taiwan, so it leaves no trade record. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes. Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

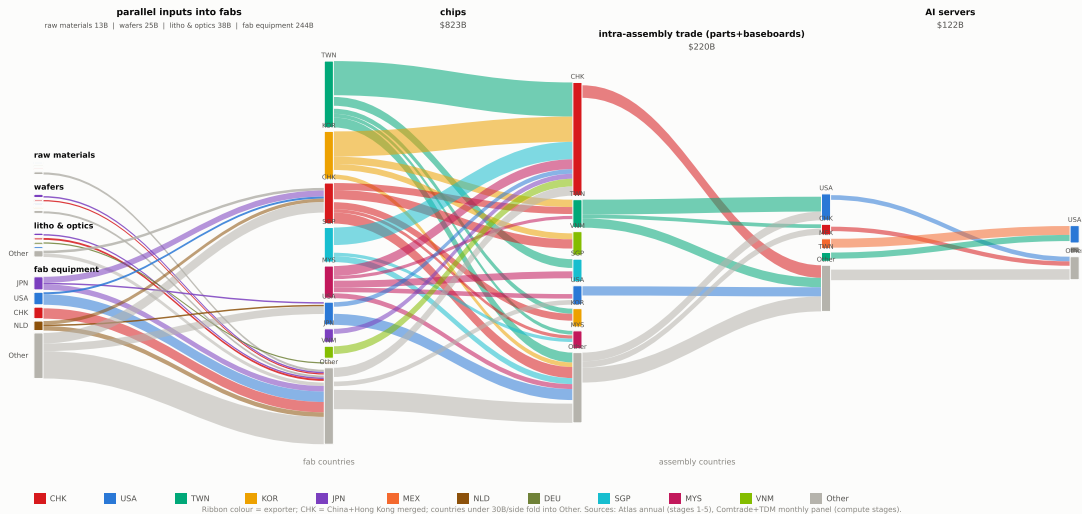
## 1. The chain as flows

# Whole-chain overview, dollar scale (coarse)

All eight stages left to right; band width = 2024 dollars.

Inputs, litho optics and equipment run in parallel and converge on the fab countries; parts and baseboards form the intra-assembly segment before servers.

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)

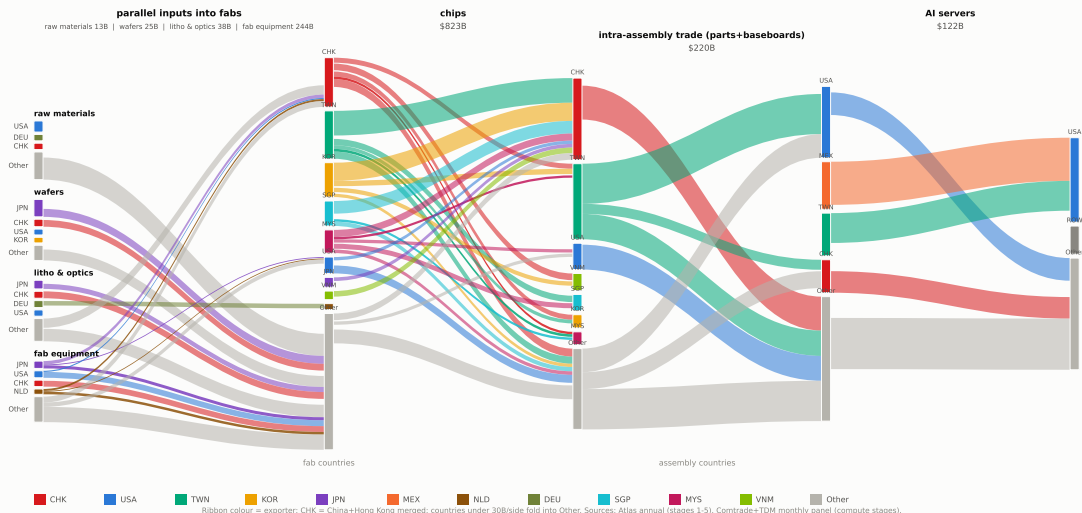


## 1. The chain as flows

# Whole-chain overview, normalized (coarse)

Same flows with each stage rescaled to equal width, so small stages' composition is readable (dollar scale makes chips dwarf everything).

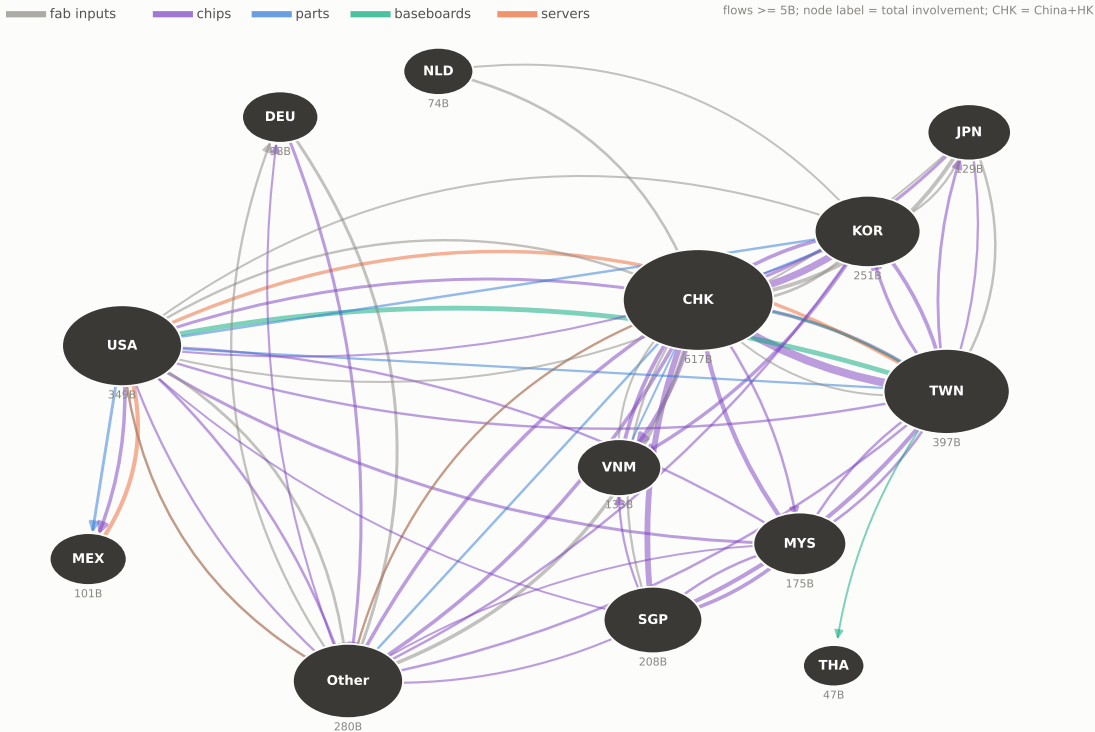
The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (stages equalized — shares only)



# Country network, all stages

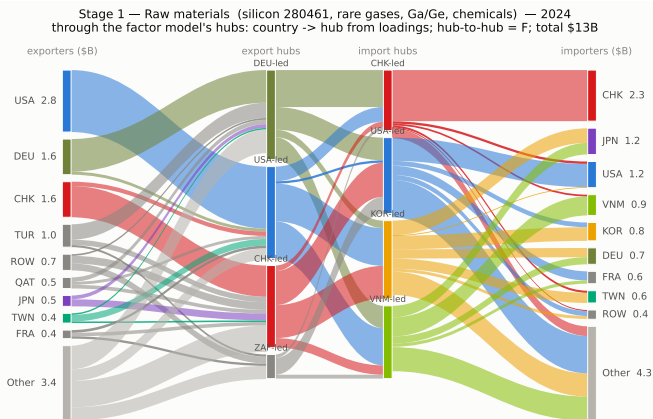
Nodes = countries, edges = 2024 flows colored by stage. Shows integrated roles a stage sequence cannot: Mexico takes in chips and parts and ships out servers at a single node.

The AI-compute supply chain as a country network, 2024 — edges by product stage, width ~ \$



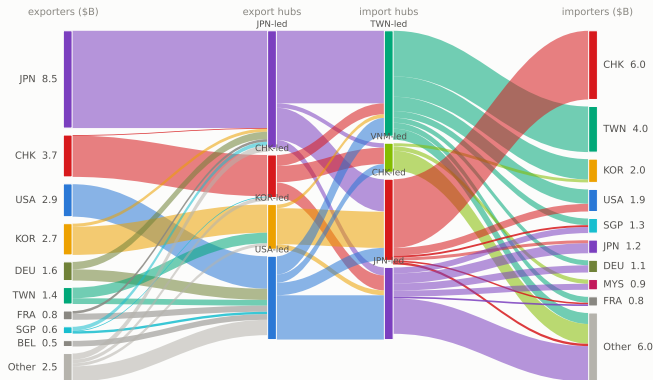
## Stage 1 -- raw materials, hub decomposition

Four-column view: exporters -> export hubs -> import hubs -> importers, hubs estimated by MFM in the nonnegativity-identified basis. Applies to every stage page that follows; fits  $R^2$  0.95-0.99.



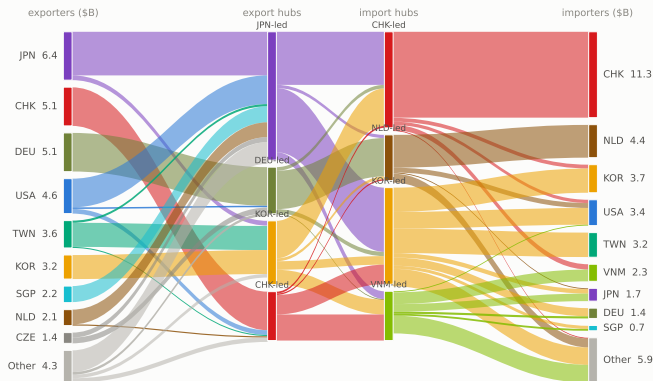
# Stage 2 -- wafers & wafer inputs

Stage 2 — Wafers & wafer inputs (381800, 3701xx/370790) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$25B



# Stage 3 -- lithography & optics inputs

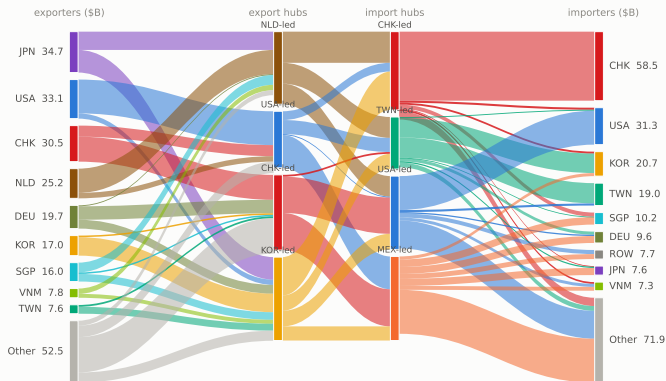
Stage 3 — Lithography & optics inputs (9001xx/9002xx, 901210/90, 903141) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$38B



## Stage 4 -- fab equipment

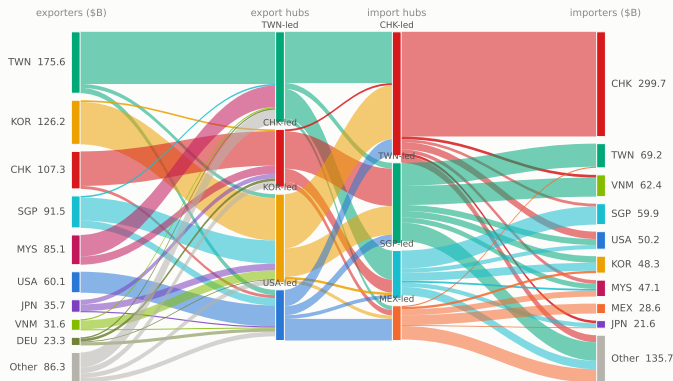
Capacity goods, not consumables: in fab countries these imports lead chip exports by 6-12 months (docs/notes/edge-type-tests.md).

Stage 4 — Fab equipment (8486xx, metrology, fab plant) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B

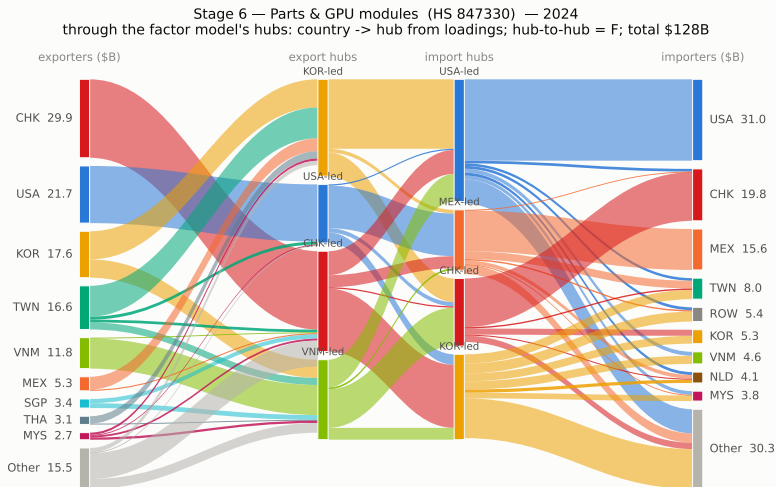


# Stage 5 -- chips (logic, memory, discretetes)

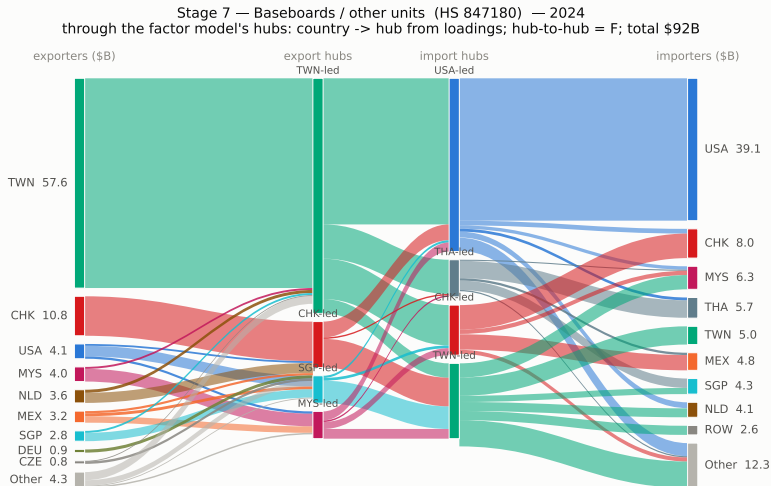
Stage 5 — Chips (logic, memory, discretetes) (8542xx, 8541xx, media) — 2024  
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



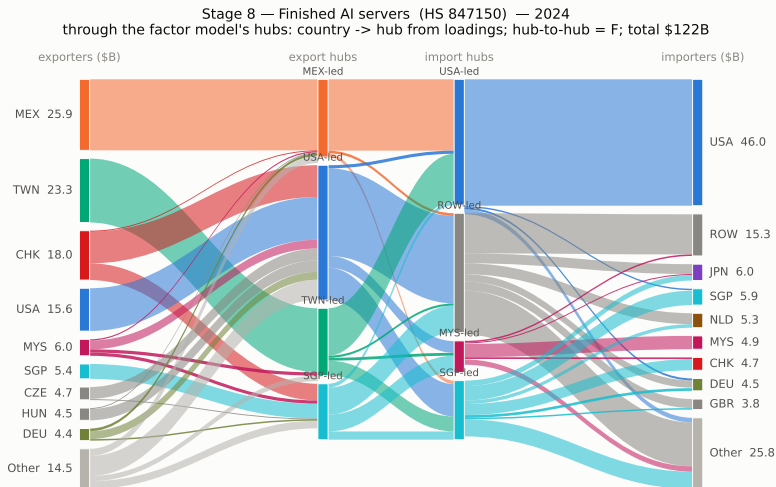
# Stage 6 -- parts & GPU modules (847330)



# Stage 7 -- baseboards (847180)



# Stage 8 -- AI servers (847150)



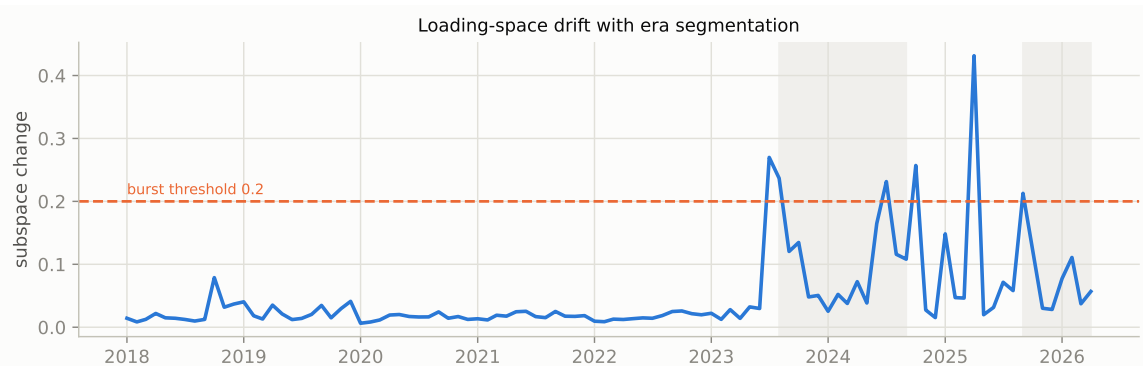
## 2. The time-varying factor model

Era-anchored TV-MFM on the audited monthly panel (12m trailing window,  $k=4$ , \$B levels, NNF basis), 2017-12..2026-04. Shaded bands = eras between structural breaks. Generated by `scripts/tvmfm_monthly_anchored.py`; summaries in `results/mfm/tvmfm/`.

2. The time-varying factor model

## AI compute, CHK basis -- loading drift and eras

The preferred basis (entrepot churn netted; Taiwan separate).  
One unbroken 68-month era from 2017-12, then breaks at  
2023-08, 2024-10, 2025-09.

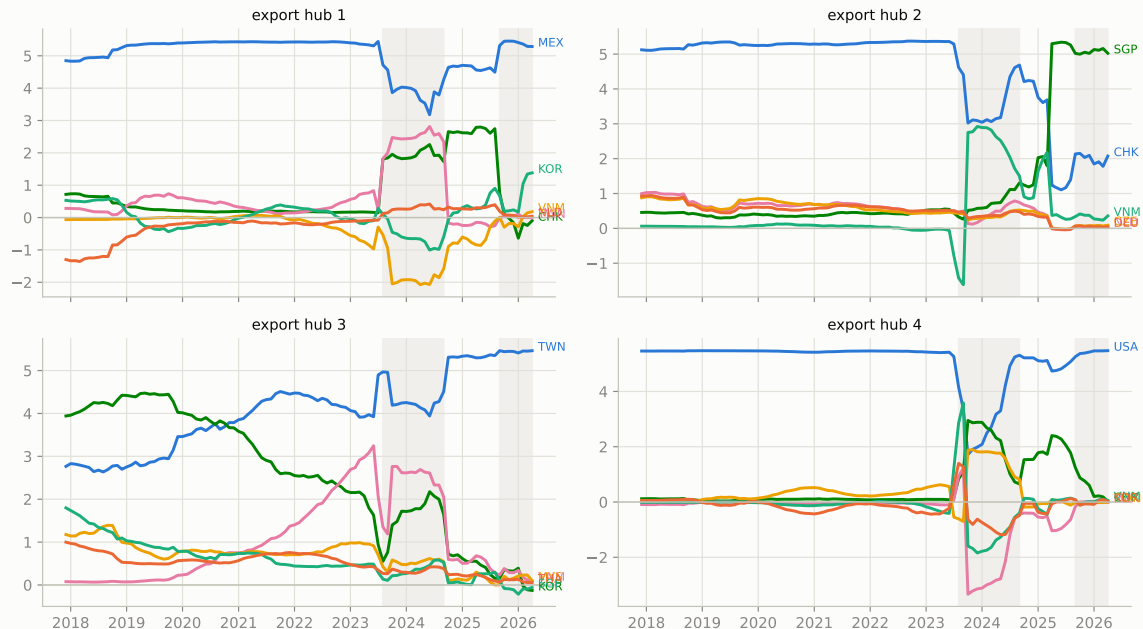


2. The time-varying factor model

## AI compute, CHK basis -- export-hub loadings

Per-hub country loadings through time. At 2024-10 the China-bloc hub loses its separate identity (cross-era similarity 0.45) and a Singapore-led hub appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

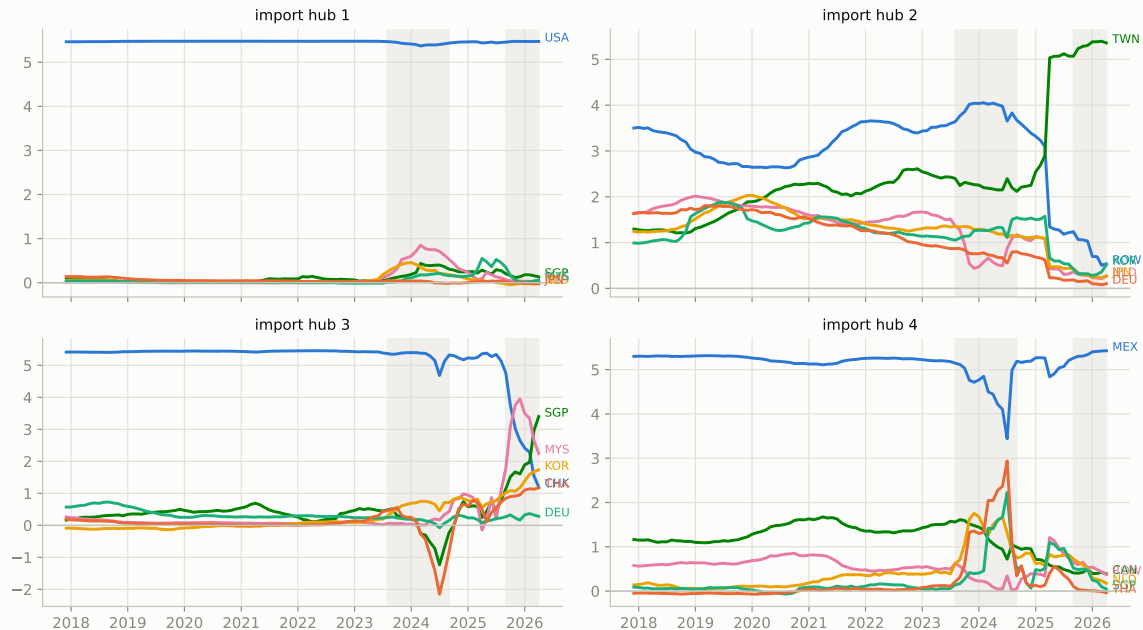


## 2. The time-varying factor model

# AI compute, CHK basis -- import-hub loadings

Import hubs are the stable side of the network (USA, Mexico, Taiwan, rest-of-world led).

AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)

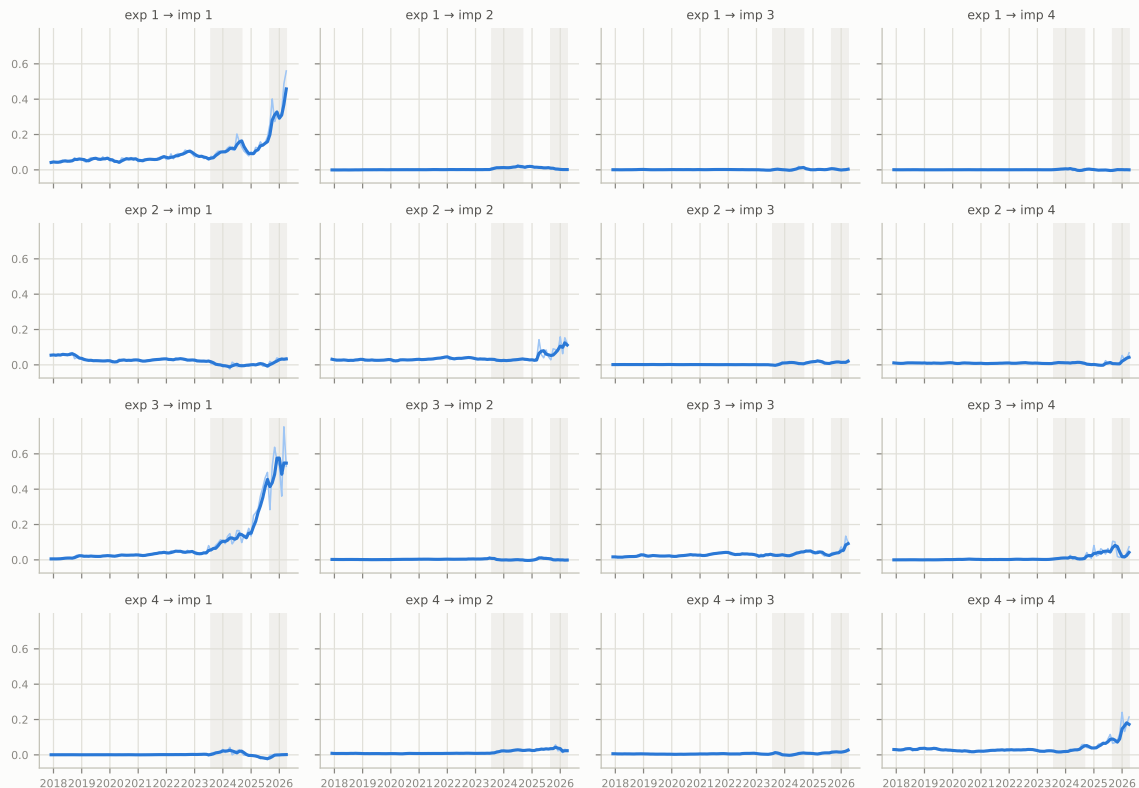


## 2. The time-varying factor model

# AI compute, CHK basis -- hub-to-hub intensities $F_t$

Each panel is one export-hub x import-hub channel in \$B (3m MA thick).

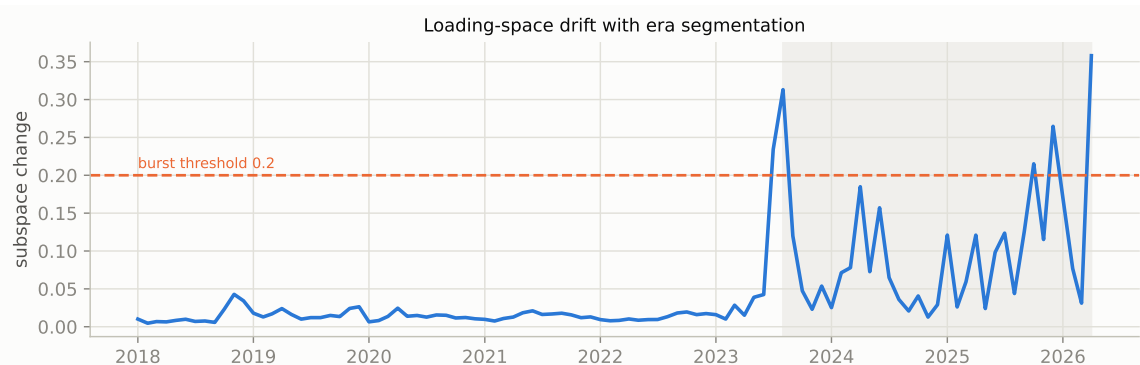
AI compute, China+HKG bloc (TWN separate) — hub-to-hub  $F_t$ , era-anchored (3m MA thick)



2. The time-varying factor model

## AI compute, country level -- drift

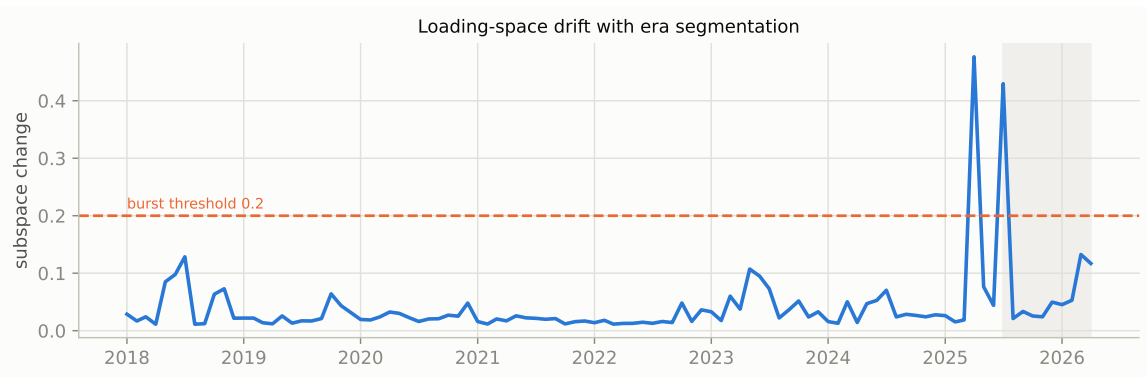
Without bloc merging the sample splits once: 2023-08. The demand event is the only dominant country-level break in nine years.



## 2. The time-varying factor model

### **Parts (847330), country level -- drift**

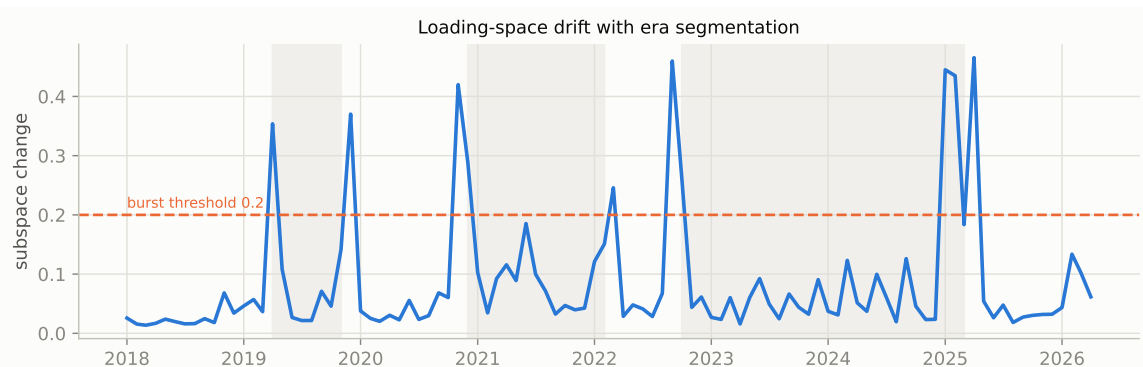
91 months without a structural break; the parts layer is the stable substrate of the network.



2. The time-varying factor model

## AI compute, double-bloc basis -- drift

CHK and USA+MEX both merged, isolating cross-bloc structure.  
Breaks at 2019-04/2019-12 (first tariff war), 2022-10 (first export controls), 2025-04 (tariffs): policy breaks blocs.

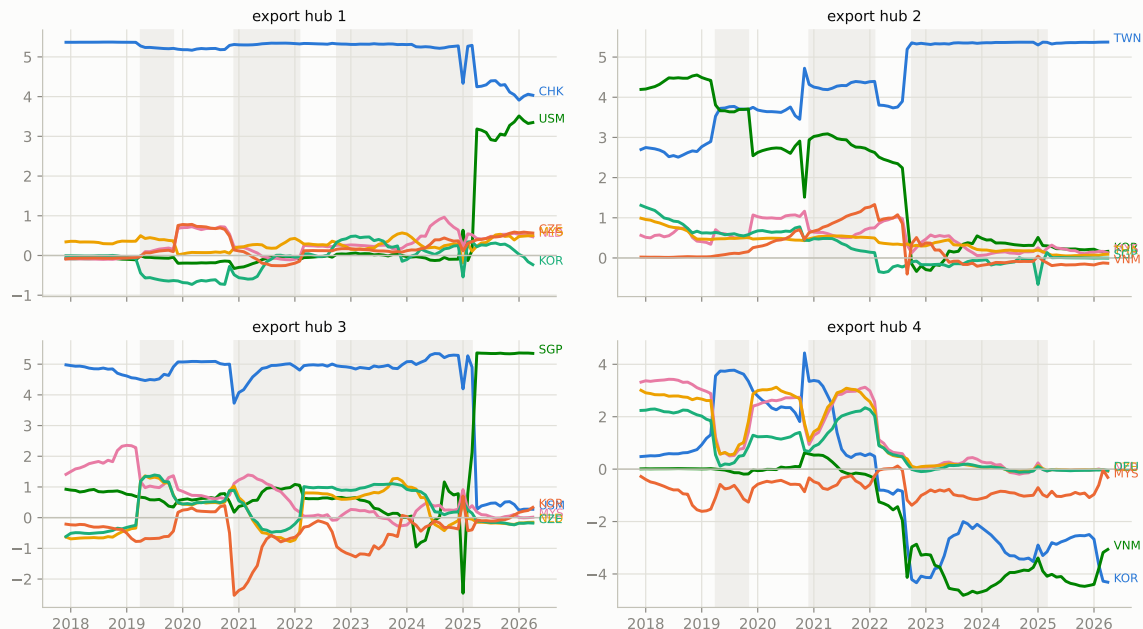


## 2. The time-varying factor model

# AI compute, double-bloc basis -- export-hub loadings

In the 2025-04 era the two blocs load together on one factor (CHK +3.97, USM +3.41) -- the entanglement signature unique to the tariff break.

AI compute, CHN+HKG and USA+MEX blocs — export loadings, era-anchored (12m window; shaded bands = eras)



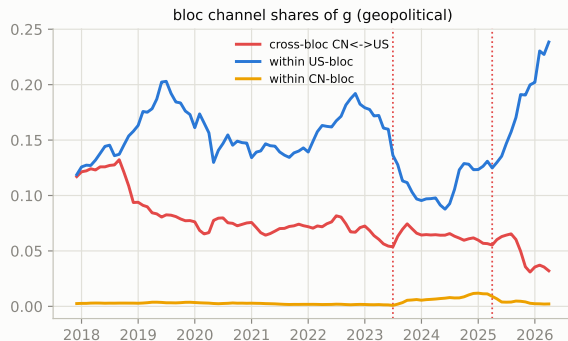
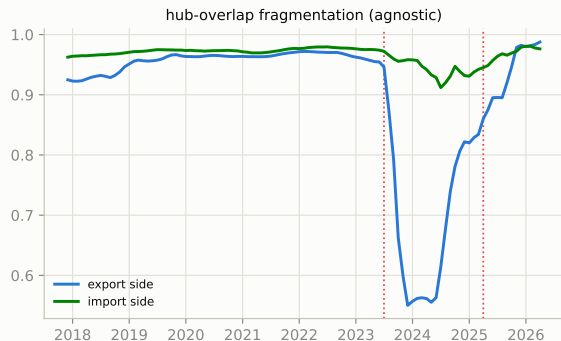
### **3. Concentration & fragmentation**

Measure system of docs/concentration-fragmentation-nnf-mfm-trade.md computed monthly from the TV-MFM (CHK basis).  
Dotted lines mark 2023-07 and 2025-04. Interpretation:  
results/network\_stats/findings.md. Generated by  
scripts/network\_stats.py.

## Two notions of fragmentation -- AI compute

Left: hub-overlap fragmentation (range-compressed under the anchored basis; mainly a diagnostic). Right: bloc channel shares -- the cross-bloc CN-US share falls ~70% over the sample in two legs (2019, then 2023 onward) with a plateau between.

AI compute (3 codes) — two notions of fragmentation (3m MA)



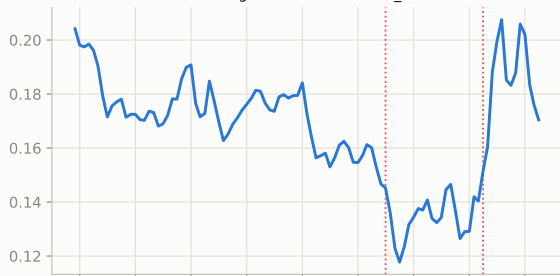
### 3. Concentration & fragmentation

## Joint concentration decomposition -- AI compute

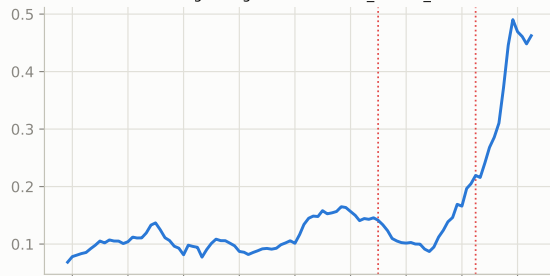
$\text{HHI\_flat} = \text{HHI\_F} \times (\mu_a \mu_b + \text{alignment covariance})$ , exact identity. Alignment covariance is mildly positive from 2017, peaks ~2019, decays through the AI era: no compounding-concentration signature.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

linkage concentration  $\text{HHI\_F}$



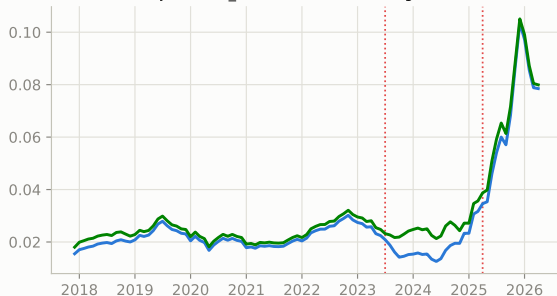
linkage-weighted means  $\mu_a * \mu_b$



alignment covariance  $\text{Cov}(a,b)$



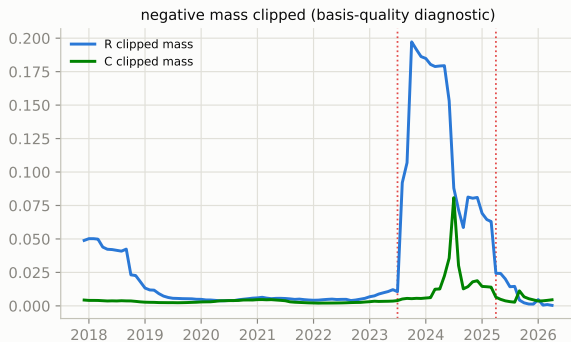
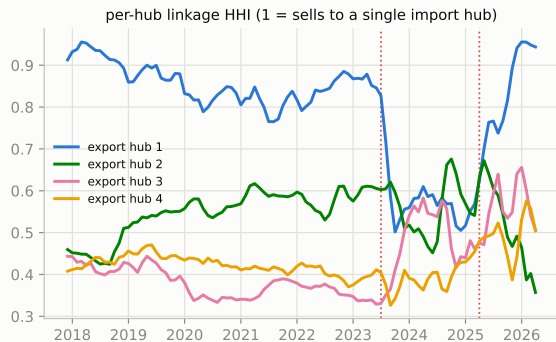
joint:  $\text{HHI\_flat}$  (blue) vs bilateral (green)



## Per-hub linkage concentration and basis diagnostic -- AI compute

Left: does an export hub sell into one import program or several. Right: clipped negative mass (basis quality), elevated during the 2023-24 regime transition.

AI compute (3 codes) — linkage structure and basis diagnostics

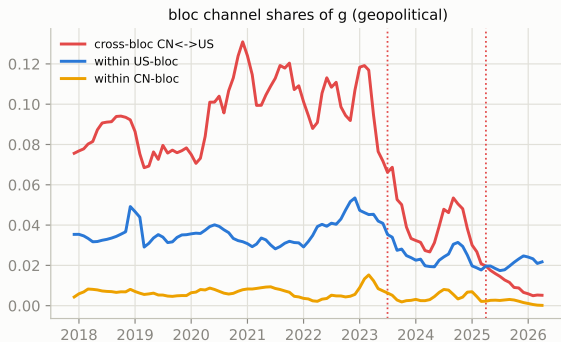
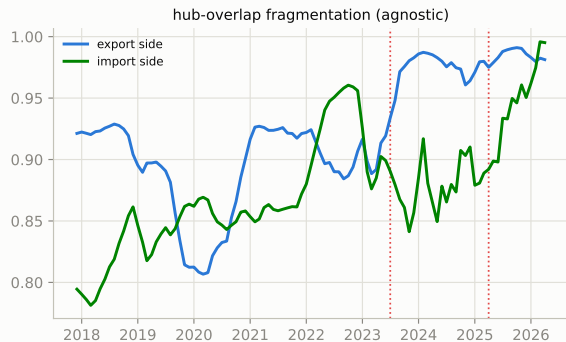


### 3. Concentration & fragmentation

## Fragmentation -- baseboards (847180)

The sharpest decoupling of the three codes: the cross-bloc share collapses ~20x from its 2021 peak while linkage concentration HHI\_F rises 0.09 -> 0.48.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)



### 3. Concentration & fragmentation

## Fragmentation -- parts (847330)

Cross-bloc share halves but remains the highest of the three codes: the parts layer stays the shared substrate.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

